UNIT I& II

1. (a) State De-morgan's Laws?
   (b) Determine the values of A, B, C, and D that make the sum term $\overline{A} + B + \overline{C} + D$ equal to zero.

   Or

   (c) Implement AND, NOT, EXOR, OR and EXNOR gate with any of the universal gates?
   (d) Solve the following Boolean Expression with its implementation using universal gates?

   AB'C+ABC+ABC'

2. Simplify the following Boolean expression using Karnaugh map: F(x,y,z)= Σ(3,4,6,7)?

1. 3. Simplify the following Boolean expression using Karnaugh map: F(x,y,z)= Σ(3,4,6,7)?

UNIT III

1. 1. Explain Full adder with its truth table, Boolean expressions, and implementation with gates?

3. Implement Full adder with Decoder?

2. Explain half Subtractor with truth table and its implementation?

3. Implement one bit comparator with its truth table?

4 Implement AND, NOT,OR gate with 2x1 Multiplexer?

5. Implement Full adder with Decoder?
6. Implement Binary adder with truth table and Boolean expression?
7 Implement EXOR, EXNOR with 2x1 Multiplexer?
8. Explain implementation of OR gate with 2X1 Multiplexer?

9. Explain half Subtractor with truth table and its implementation?

10.Implement one bit comparator with its truth table?

UNIT IV

1. Explain SR-Latch with True table.?
2. What is the difference between a Mealy and Moore State Machine.?

1. Explain SR-Flip Flop with State table and State diagram?
2. Draw the T flip-flop using NAND gates with a function table?

1. 2. Implement two-bit Comparator with its truth table and Boolean expression?

3. Implement Full subtractor with a decoder?

4 Implement 8x1 Multiplexer with the help of 2x1 multiplexer?

5. What is decade counter and explain its working.?
6. Explain JK-Flip Flop with function table, characteristics table and Excitation table.?

7. Explain PIP and PISO with example?

a. Explain the 4-bit Johnson counter?
b. Explain the JK flip-flop with a state table and a state diagram?

(Or)

c. Explain T-Flip Flop with function table, characteristics table and Excitation table.?
d. Explain shift registers.?

UNIT V

1. What is Hamming code and how does it work?\

2. How does Hamming Code function, and how does it identify errors in an 8-bit data word such as 11010101?

3. Tabulate the PLA programming table for three Boolean functions listed below. Minimize the number of product terms

X= AB'+AB'C+BC
Y= (AB'C'+AB'+B'C')
Z= (AB'+A'B'C'+AC'+BC)

Or

4.Tabulate the PLA programming table for four Boolean functions listed below. Minimize the number of product terms

A(x,y,z)=∑(0,1,2,5,7)
B(x,y,z)=∑(0,3.2,4,5,6)

5. Tabulate the PLA programming table for three Boolean functions listed below. Minimize the number of products terms

X= AB'+A'C+A'BC
Y= (A'C'+B'C'+ABC')

6.Tabulate the PAL programming table for three Boolean functions listed below. Minimize the number of products terms.

X= AB'+ABC+A'BC
Y= (AB'C'+AB'+A'B'C')
z= (AB'C'+A'B'C'+ABC'+A)

7. Tabulate the PAL programming table for three Boolean functions listed below. Minimize the number of product terms

X= AB'+ABC+A'BC
Y= (AB'C'+AB'+A'B'C'+AB)
Z= (AB'+A'B'C'+ABC'+BC)

8. Tabulate the PAL programming table for four Boolean functions listed below. Minimize the number of products terms.

A(x,y,z)=∑(0,1,5,7)
B(x,y,z)=∑(2,4,5,6)
C(x,y,z)=∑(0,1,2,3,4)
D(x,y,z)=∑(3,6,7)

9. Explain Static RAM and Dynamic RAM?

10. Explain types of ROMs?

2. (a) Convert $(128)_{10}$ to octal number?
   (b) Convert $(128)_{10}$ to hex number?
3. Simplify the Boolean function F= A + A′B. How many gates will be required to implement this expression?
4. Simplify the following Boolean expression using Karnaugh map: F(x,y) = x'y+x'y'?
5. Explain SR-Latch with True table.?
6. What is the difference between a Mealy and Moore State Machine.?

2.
   a. Explain the 4-bit Johnson counter?
   b. Explain the JK flip-flop with a state table and a state diagram?

   (Or)

   c. Explain T-Flip Flop with function table, characteristics table and Excitation table.?
   d. Explain shift registers.?

3.
   a. Tabulate the PLA programming table for three Boolean functions listed below. Minimize the number of products terms

   X= AB'+A'C+A'BC

Y= (A'C'+B'C'+ABC')

b. How does Hamming Code function, and how does it identify errors in an 8-bit data word such as 11110101?

Or

c. Tabulate the PAL programming table for three Boolean functions listed below. Minimize the number of products terms.

X= AB'+ABC+A'BC

Y= (AB'C'+AB'+A'B'C')

z= (AB'C'+A'B'C'+ABC'+A)

d. Tabulate the PAL programming table for four Boolean functions listed below. Minimize the number of products terms.

$A(x,y,z)=\sum(0,1,5,7)$

$B(x,y,z)=\sum(2,4,5,6)$

$C(x,y,z)=\sum(0,1,2,3,4)$

$D(x,y,z)=\sum(3,6,7)$